

Differentiation & Trig Identities — Jargon & Formula Recall Pack

Harry Williams → Phuc Nguyen · Tutorial of 12 July 2026 · Edexcel IAL Pure Mathematics P3 (WMA13/01) · applied to the June 2024 P3 paper

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Contents

0. How to read this pack	2
0.1 Status labels (used throughout)	2
0.2 The four boundaries this pack keeps explicit	3
1. Jargon glossary	3
2. Exponential / log / power / product / quotient / chain derivatives	6
3. Reciprocal & standard trig derivatives	7
4. Trig identities — reciprocal / Pythagorean / double-angle	8
4.1 Reciprocal identities (C1-C4) [T]	8
4.2 Pythagorean identities (C5-C7) [T]	8
4.3 Double-angle identities (C8-C9) [T]	8
5. R-form & solving the trig equation (Q4)	9
5.1 Q4(a) — to $a \sin 2x + b \cos 2x + c$ · Solved	9
5.2 Q4(b) — to $R \sin(2x + \alpha) + c$, $R > 0$, $0 < \alpha < \frac{\pi}{2}$ · Solved	9
5.3 Q4(c) — max value and second smallest x	10
6. Transformations & integration applications (Q1, Q2)	10
6.1 Q1 — modulus & transformations · $f(x) = 2 x - 5 + 10$, P the vertex	10
6.2 Q2 — algebraic division, then integration · $g(x) = \frac{2x^2 - 5x + 8}{x - 2}$	11
6.3 Q3 — log-log graph · Discussed only (D)	11

7. Evidence-based Phuc traps & technique boundaries	11
7.1 Phuc's own errors → Harry's corrections (diagnostic) [T]	11
7.2 The four explicit boundaries (consolidated)	12
7.3 Exam-technique boundaries Harry named [T]	12
7.4 Progress / study-plan context (not taught maths) [T]	12
7.5 Independent-study advice — Harry's email (email, not video) [E]	13
8. Sources & status boundaries (compact)	13

0. How to read this pack

This is a **recall pack**: dense, cross-tabulated definitions and formula/method tables for the 12 July 2026 tutorial, meant for fast pre-exam review and for a native-Notion table export. It condenses the source-grounded parent guide (guide/harry_2026-07-12_differentiation_trig_revision_guide.tex/.pdf) and its evidence_coverage.md into look-up tables. Nothing here is invented; every row traces to one of three source tiers, tagged so you can tell **tutor voice** from **official text**:

- **[T] Tutor-taught** — what Harry actually said or wrote in the Loom recording: the spoken transcript sources/loom_transcript.txt, corroborated by the six-segment audio+visual ledger analysis/gemini_video_evidence.md (the governing evidence). This is the primary evidence and is never overwritten. **The June 2024 P3 question statements below are [T]** — they are reconstructed from Harry's shared board, because **no verbatim question-paper text file was staged** for this lesson (contrast the sibling P3-Oct-2025 pack, which had a [Q] paper). Treat the question wording as *tutor-transcribed*, not official.
- **[S] Official / spec-derived reference** — from the *Edexcel IAL Mathematics 2018 specification* (sources/Edexcel_IAL_Maths_2018_specification.txt) and its formula booklet. Used **only** to expand the formal statement behind a method: the standard **domain conditions** ($a > 0$, $f(x) > 0$, $\cos A \neq 0$), the **radian assumption** for the derivative formulae, the algebraic **derivations** of the Pythagorean identities, and the general **R -formula** / $\sin \theta = 1$ **solution rules**. These are always labelled **[S]** and are **never** put into Harry's mouth. In the parent guide they carry a dagger ([†], §8).
- **[E] Email** — Harry's follow-up email of 12 July 2026 (sources/harry_followup_email_2026-07-12.txt). Independent-study / spaced-practice advice; **email guidance, not video teaching** (§8).

SaveMyExams HTML pages (sources/savemyexams_*.html) were consulted as reference cross-checks only and were **not** allowed to overwrite tutor / spec evidence.

0.1 Status labels (used throughout)

Label	Meaning
S — Solved	A correct final answer was reached in the lesson.
P — Partially solved	Carried forward mid-solution.
D — Discussed only	Talked through without a full written solution.

Load-bearing status boundaries for this lesson (repeated in §7–§8):

- **Q3 (log-log graph) = discussed only (D)**. Phuc's sketch was "pretty much fine"; no full tutor working exists, so none is reconstructed here.
- **Everything else drilled (warm-up A, trig table B, identities C, Q1, Q2, Q4) = solved (S)** on the board.
- The **progress review / study-plan** talk (score, readiness, scheduling) is **context, not taught maths** — only the maths-relevant slivers are kept (§7.4).

0.2 The four boundaries this pack keeps explicit

Because these are the exact places marks (and provenance) leak, each has its own flagged treatment and is called out inline wherever it applies:

1. $f(x) > 0$ — validity condition on $\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}$ and on dropping the modulus in $\ln |f(x)|$. **[S]-derived** (a mathematical validity condition; all lesson examples satisfy it), *not* a verbatim tutor statement. See §2 and §6.
2. **Radians** — the derivative formulae and the R -form angle assume x/α in **radians**; Q4(b) explicitly wanted α in radians to 3 s.f. The radian *assumption on the derivative formulae* is **[S]**; the Q4 radian *demand* is **[T]** (from the board). See §7.2.
3. **Phase / quadrant** — in the R -form, $\tan \alpha = \frac{b}{a}$ alone does **not** fix α ; the quadrant is fixed by the **signs of** a, b (equivalently $\alpha = \text{atan2}(b, a)$). **[S]-derived**. Here $a, b > 0 \Rightarrow$ first-quadrant α . See §5.
4. **Tutor-vs-official derived boundary** — every [S] row is a specification expansion added for consolidation, **not** something Harry said. The tutor drilled the *formulae from memory*; the *conditions and general rules* around them are the official layer. This boundary is the spine of §0 and §8.

1. Jargon glossary

The canonical 7-column recall table. Definitions are 30–80 words. The final column is a **Phuc trap** only where the recording shows Phuc actually making that error; otherwise it is a **Common trap** (a standard pitfall Harry warned about, or an examiner convention). Tier tags: [T] tutor, [S] spec, [E] email.

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
Power rule [T][S]	To differentiate a variable raised to a <i>fixed number</i> , drop the power to the front and knock one off it.	$\frac{d}{dx}(x^n) = n x^{n-1}$. The variable is in the base , the constant is the exponent.	n constant; variable in the base.	Not the a^x rule. If the variable is in the <i>power</i> , this rule does not apply.	$\frac{d}{dx}(x^3) = 3x^2$, $\frac{d}{dx}(x^4) = 4x^3$ (basics check).	Phuc trap: first wrote $\frac{d}{dx}(a^x) = x a^{x-1}$ — power rule mis-applied to an exponential. Harry: check where the variable is .
Exponential-base rule a^x [T][S]	Differentiating a <i>number</i> to a variable power leaves the number-power alone and pulls down a natural log of the base.	$\frac{d}{dx}(a^x) = a^x \ln a$. Variable in the power ; base a constant.	$a > 0$ constant [S] . Chain form: $\frac{d}{dx}(a^{kx}) = k a^{kx} \ln a$.	Not $x a^{x-1}$ (power rule) and the factor is $\ln a$, not $\ln(kx)$.	$\frac{d}{dx}(3^x) = 3^x \ln 3$; $\frac{d}{dx}(3^{2x}) = 2 \cdot 3^{2x} \ln 3$; $\frac{d}{dt}(9^t) = 9^t \ln 9 = 2 \cdot 9^t \ln 3$.	Phuc trap: the $x a^{x-1}$ slip above. Harry: " <i>a is a constant, x is a variable</i> " — variable in the power $\Rightarrow a^x \ln a$.

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
In $f(x)$ derivative (chain) [T][S]	Differentiate a natural log by putting the derivative of the inside on top and keeping the inside on the bottom.	$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}$.	Requires $f(x) > 0$ so $\ln f(x)$ is real [S] (validity condition; all lesson examples satisfy it).	Not just $\frac{1}{f(x)}$ — you must include $f'(x)$ on top.	$\frac{d}{dx} \ln(x^2 + 3x + 5) = \frac{2x + 3}{x^2 + 3x + 5}$; for $\ln(kx)$ the k cancels : $\frac{k}{kx} = \frac{1}{x}$.	Phuc trap : wrote $\frac{1}{3x}$ for $\frac{d}{dx} \ln 3x$ (dropped $f' = 3$). Harry: “differentiate the argument, and that goes on the top.”
Constant multiple / neatness [T]	A constant just rides along; by convention write it at the front.	$\frac{d}{dx}(c g(x)) = c g'(x)$. Harry’s convention: constants at the front.	c constant.	Not a term you differentiate; it does not gain a power or a log.	$\frac{d}{dx}(5 \ln x) = \frac{5}{x}$; $\frac{d}{dx}(5 \cdot 4^x) = 5 \cdot 4^x \ln 4$.	Common trap : Harry (05:47): “we tend to put these constants at the front ... not wrong the other way, but it just looks a little odd.”
Reciprocal trig functions [T][S]	The “co-” partners: cosec, sec, cot are 1 over sin, cos, tan.	$\csc A = \frac{1}{\sin A}$, $\sec A = \frac{1}{\cos A}$, $\cot A = \frac{1}{\tan A} = \frac{\cos A}{\sin A}$.	Undefined where the denominator is 0 [S] ($\sin A = 0$, $\cos A = 0$, resp.).	sec is not $1/\sin$. Use the third letter : cosec $\rightarrow$ sin, sec $\rightarrow$ cos.	Recall drill: Phuc gave all four correctly.	Common trap : pairing sec with sin. Harry’s hook: “cosec is one over sine ... the third letter, remember?”
Standard trig derivatives [T][S]	The six memorised derivatives, with an alternating sign pattern.	$\sin \rightarrow \cos$, $\cos \rightarrow -\sin$, $\tan \rightarrow \sec^2$, $\csc \rightarrow -\csc \cot$, $\sec \rightarrow \sec \tan$, $\cot \rightarrow -\csc^2$.	Assume radians [S] . Order sin, cos, tan, csc, sec, cot gives signs + − + − + −.	Not reprinted in the exam formula booklet — you must know them.	Board table; $\tan \rightarrow \sec^2$ links to $1 + \tan^2 = \sec^2$.	Phuc trap : wrote $\cos x$ with no sign, and “ $\pm$ ” for $\frac{d}{dx} \cos x$: “I usually confuse about the plus minus.” Fix: the + − + − + − pattern.
Pythagorean identities [T][S]	The “ $1 = \sin^2 + \cos^2$ ” family and its two reciprocal cousins.	$\sin^2 A + \cos^2 A = 1$; $1 + \tan^2 A = \sec^2 A$; $1 + \cot^2 A = \csc^2 A$.	The two cousins come from dividing the master identity by $\cos^2 A$ / $\sin^2 A$ [S] .	$1 + \tan^2 A$ is not $\cot^2 A$.	Recall drill C5-C7; feeds $4 \cos^2 x$ in Q4(a).	Phuc trap : answered $\cot^2 A$ for $1 + \tan^2 A$. Harry: it is $\sec^2 A$ (same pair as $\frac{d}{dx} \tan x = \sec^2 x$).
Double-angle identities [T][S]	Formulas for sin/cos of $2A$ in terms of A ; memorise the three $\cos 2A$ forms.	$\sin 2A = 2 \sin A \cos A$; $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$.	Order in $\cos 2A$: cosine first . Derivations are [S] (for understanding).	$\cos 2A$ is not $\sin^2 A - \cos^2 A$ and not $\cos^2 A - 2 \sin^2 A$.	Q4(a): $2 \cos^2 A - 1$ rearranges to $2 \cos^2 A = \cos 2A + 1$.	Phuc trap : reached for $\sin(A + B)$, then wrote wrong order/forms. Harry: “memorise these three forms rather than deriving them during the exam” $\rightarrow$ flashcards.
R-form (harmonic / auxiliary angle) [T][S]	Rewrite “ $a \sin + b \cos$ ” as one shifted sine wave of amplitude R .	$a \sin \theta + b \cos \theta = R \sin(\theta + \alpha)$; match $R \cos \alpha = a$, $R \sin \alpha = b$.	$R > 0$; $R = \sqrt{a^2 + b^2}$ exact; α fixed by the signs of a, b (quadrant), not by $\tan \alpha$ alone [S] ; radians here.	R is not a decimal when “exact” is asked. $\tan \alpha = \frac{b}{a}$ alone does not fix α .	Q4(b): $4 \sin 2x + 2 \cos 2x = 2\sqrt{5} \sin(2x + \alpha)$, $\alpha = \arctan \frac{1}{2} \approx 0.464$.	Phuc trap : wrote $R = 4.472$. Harry: “we want the exact value of R ... no decimals allowed” $\Rightarrow R = 2\sqrt{5}$.
Exact value [T][S]	Leave surds / fractions / π — no rounded decimals.	A value with no rounding: $\sqrt{20} = 2\sqrt{5}$, $\arctan \frac{1}{2}$, $2\sqrt{5} - 1$ kept symbolic.	Applies whenever the command word says “exact”.	Not a decimal ($R = 4.472$ is <i>not</i> exact).	$R = 2\sqrt{5}$; $f_{\max} = 2\sqrt{5} - 1$.	Phuc trap : decimal $R = 4.472$. Harry: “Exact, no decimals allowed.”

Term	Student-friendly meaning	Exact mathematical meaning	Symbols / conditions	What it is NOT	Lesson example	Phuc / common trap
Significant figures ≠ decimal places [T]	"3 s.f." counts from the first non-zero digit; "3 d.p." counts after the point.	3.694 to 3 s.f. is 3.69 (not 3.694, which is 4 s.f. / 3 d.p.).	Round the <i>final</i> answer to the <i>demand</i> ed accuracy.	s.f. is not d.p.	Q4(c)(ii): $3.694 \rightarrow 3.69$ (3 s.f.).	Phuc trap: quoted 3.694. Harry: $3 \text{ s.f.} \neq 3 \text{ d.p.}$
Whole-angle periodicity [T][S]	When you solve $\sin(\text{angle}) = 1$, add the 2π to the <i>angle</i> , before dividing to get x .	$\sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2} + 2n\pi$ [S]; the 2π belongs to $\theta = 2x + \alpha$, not to x .	Solve for the whole angle first , then rearrange for x .	The x -values are not 2π apart after you divide by 2.	Q4(c)(ii): $2x + \alpha = \frac{\pi}{2}, \frac{5\pi}{2} \Rightarrow x \approx 0.553, 3.69$.	Phuc trap (headline): added 2π to x . Harry: "these ones, when you've rearranged, are not two π apart."
Modulus function [T][S]	A "V"-shaped graph; the vertex is the corner.	$f(x) = a x - h + k$ has vertex (h, k) ; split into two linear branches to solve algebraically.	Piecewise: $2 x - 5 + 10 = 2x$ for $x \geq 5$, $= -2x + 20$ for $x < 5$.	Not smooth at the vertex; "calculator solutions not acceptable" on Q1(b).	Q1: vertex $P = (5, 10)$; solve $2 x - 5 + 10 > 6x \Rightarrow x < \frac{5}{2}$.	Common trap: solving the wrong branch — sketch first to pick the branch, then do algebra.
Function transformation [T][S]	$a f(x - h)$: inside shifts x , outside scales y .	$y = 3f(x - 2)$: $x \rightarrow x + 2$ (right 2), $y \rightarrow 3y$ (stretch $\times 3$) applied to each point.	Inside operations act on x (opposite sense); outside on y .	The stretch factor multiplies y , not x .	Q1(c): $P = (5, 10) \rightarrow P' = (7, 30)$.	Common trap: mixing which coordinate moves — inside $\Rightarrow x$, outside $\Rightarrow y$.
Algebraic (polynomial) division [T][S]	Divide top by bottom to split a fraction into a polynomial plus a remainder term.	$\frac{2x^2 - 5x + 8}{x - 2} = 2x - 1 + \frac{6}{x - 2}$ (quotient $2x - 1$, remainder 6).	Remainder sits over the divisor: $\frac{6}{x - 2}$, not $(x - 2)(2x - 1) + 6$ over $(x - 2)$.	Not the same as factorising; there is a leftover $\frac{C}{x - 2}$.	Q2(a): $A = 2, B = -1, C = 6$; also by substitution/identity.	Phuc trap: $-5x - (-4x) = -9x$. Harry: "always bracket the term you subtract" $\Rightarrow -x$.
$\int \frac{1}{x} / \int \frac{f'}{f}$ (log integral) [T][S]	Integrating "derivative over function" gives a natural log.	$\int \frac{1}{x} dx = \ln x + c$; $\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$ (reverse of the $\ln f(x)$ derivative).	Keep the modulus ; drop it only where the argument is known positive ($f(x) > 0$) [S].	Not a power-rule integral; there is no $\frac{x^0}{0}$.	Q2(b): $\int \frac{6}{x - 2} = 6 \ln x - 2 $; on $[4, 8], x - 2 > 0$.	Common trap: forgetting the modulus, or the log law $\ln 6 - \ln 2 = \ln \frac{6}{2} = \ln 3$.
Command word "sketch" vs "plot" [T]	A <i>sketch</i> needs shape + key points; a <i>plot</i> needs an accurate scale.	"Sketch": show shape and label intercepts as coordinate pairs. "Plot": accurate scaled graph.	Label intercepts as pairs, e.g. $(0, 6), (2, 0)$.	A sketch is not graph paper to scale.	Q3 (discussed only): Phuc's $\log_{10} y$ -vs- $\log_{10} x$ sketch "pretty much fine".	Common trap: giving un-labelled intercepts; use coordinate pairs.
"Invent-and-carry" exam tactic [T]	If you're stuck on an early part, make up a plausible value and carry it forward.	Assume a value for a missing early result so later parts still earn method marks (carry-forward).	Later parts marked on method, not on the (wrong) carried value.	Not a licence to guess the <i>final</i> answer; it rescues <i>method</i> marks.	Harry stressed it $3 \times$ (Q1).	Common trap: leaving later parts blank because an early part failed.
Forgetting curve / spaced practice [E]	We forget over time; spacing and revisiting beats cramming.	Space practice out; revisit learned material regularly to combat the forgetting curve.	Email guidance, not video teaching.	Not a one-and-done: re-doing seen questions is valid consolidation, though less diagnostic than an unseen paper.	Email attachments show a forgetting-curve diagram.	Common trap (per Harry): assuming re-done questions are worthless — "there is a place for both."

2. Exponential / log / power / product / quotient / chain derivatives

Warm-up rows **A1-A19** (all **Solved**). Harry opened: “*a is a constant, x is a variable.*” The product/quotient rows are **[S] official reference** — P3 content (spec 4.2) but **not separately drilled in this lesson**; kept here for a complete cross-tabulation.

Function	Derivative	Rule / condition	Trigger (when to use)	Trap	Src
x^n	$n x^{n-1}$	Power rule — variable in the base.	Variable is the base, exponent fixed.	Do not use for a^x — check where the variable sits.	[T][S]
a^x	$a^x \ln a$	$a > 0$ [S] ; variable in the power.	A number to a variable power.	Not $x a^{x-1}$; factor is $\ln a$.	[T][S]
a^{kx}	$k a^{kx} \ln a$	Chain rule pulls out the inner k .	a^{kx} , k constant.	Factor is $k \ln a$, not $\ln(kx)$.	[T]
e^x	e^x	Self-derivative (anchor case, $\ln e = 1$).	Base e .	—	[T]
e^{kx}	$k e^{kx}$	Special case of a^{kx} , $a = e$.	Base e , inner kx .	Bring the k down.	[T]
$\ln x$	$\frac{1}{x}$	Standard log derivative.	In of the variable.	—	[T][S]
$\ln f(x)$	$\frac{f'(x)}{f(x)}$	Requires $f(x) > 0$ [S] . “Argument on top.”	In of a function.	Include $f'(x)$; for $\ln(kx)$ the k cancels to $\frac{1}{x}$.	[T][S]
$c g(x)$	$c g'(x)$	Constant multiple; write c at the front.	A constant factor.	The constant doesn't gain a power/log.	[T]
$u(x) v(x)$	$u'v + uv'$	Product rule [S] — <i>not drilled this lesson</i> ; P3 spec 4.2.	Two functions multiplied.	Both terms; don't differentiate factor-by-factor separately.	[S]
$\frac{u(x)}{v(x)}$	$\frac{u'v - uv'}{v^2}$	Quotient rule [S] — <i>not drilled this lesson</i> ; spec 4.2. Minus sign, order matters.	Top-over-bottom function.	Keep the minus in the numerator; v^2 underneath.	[S]
$\text{outer}(f(x))$	$\text{outer}'(f) \cdot f'(x)$	Chain rule — differentiate outer, times inner derivative.	A function of a function.	Don't forget the inner $f'(x)$ (the k in a^{kx} , the f' in $\ln f$).	[T][S]

Worked warm-up drill (A3-A19, all Solved) [T]

#	Differentiate	Answer	Rule / note
A3	3^x	$3^x \ln 3$	a^x rule, $a = 3$.
A4	$3x^2$	$3x^2$	Power rule — contrast with a^x .
A5	$\ln 3x$	$\frac{3}{3x} = \frac{1}{x}$	Chain; $f' = 3$ on top, then k cancels.
A7	$\ln(x^2 + 3x + 5)$	$\frac{2x + 3}{x^2 + 3x + 5}$	Phuc's worked example of $\frac{f'}{f}$; confirmed.
A8	3^{2x}	$2 \cdot 3^{2x} \ln 3$	a^{kx} , $a = 3$, $k = 2$.
A9	2^{3x}	$3 \cdot 2^{3x} \ln 2$	a^{kx} , $a = 2$, $k = 3$.

#	Differentiate	Answer	Rule / note
A10	$5 \ln x$	$\frac{5}{x}$	Constant $\times \frac{1}{x}$.
A11	e^x	e^x	Self-derivative.
A12	$5 \ln 2x$	$\frac{5}{x}$	$5 \cdot \frac{2}{2x} = \frac{5}{x}$.
A13	9^t	$9^t \ln 9 = 2 \cdot 9^t \ln 3$	Variable t ; $\ln 9 = 2 \ln 3$.
A14	$5 \cdot 4^x$	$5 \cdot 4^x \ln 4$	Constant $\times a^x$.
A15	x^4	$4x^3$	Power rule (basics check).
A16	$\ln 6x$	$\frac{6}{6x} = \frac{1}{x}$	Chain; simplifies.
A17	$6 \ln x$	$\frac{6}{x}$	Constant $\times \frac{1}{x}$.
A18	$3e^{2x}$	$6e^{2x}$	$3 \cdot 2 \cdot e^{2x}$.
A19	e^{-x}	$-e^{-x}$	e^{kx} , $k = -1$.

[T] The trap Harry corrected first: Phuc's opening $\frac{d}{dx}(a^x) = x a^{x-1}$ was the *power rule* mis-applied. Fix: **variable in the base** $\Rightarrow$ **power rule; variable in the power** $\Rightarrow a^x \ln a$. **[S] $f(x) > 0$ boundary:** the $\ln f(x)$ rule needs $f(x) > 0$ for $\ln f(x)$ to be real — a validity condition, added from the spec, not a spoken lesson line.

3. Reciprocal & standard trig derivatives

Board table **B1-B8**, all **Solved**. Written in the order sin, cos, tan, csc, sec, cot the derivative signs read + − + − + −. **[S] radian assumption:** these formulae assume x in radians.

y	$\frac{dy}{dx}$	Sign	Memory hook / link	Src
$\sin x$	$\cos x$	+	Pattern position 1.	[T][S]
$\cos x$	$-\sin x$	−	Position 2 — the sign Phuc kept forgetting.	[T][S]
$\tan x$	$\sec^2 x$	+	Mirrors $1 + \tan^2 x = \sec^2 x$.	[T][S]
$\csc x$	$-\csc x \cot x$	−	Position 4; required P3 result.	[T][S]
$\sec x$	$\sec x \tan x$	+	Mirror of the csc derivative.	[T][S]
$\cot x$	$-\csc^2 x$	−	Mirrors $1 + \cot^2 x = \csc^2 x$.	[T][S]
—	Sign pattern + − + − + −		"a really nice pattern ... plus, minus, plus, minus, plus, minus."	[T]
—	Derivative $\leftrightarrow$ identity link		tan $\rightarrow$ sec ² pairs with $1 + \tan^2 = \sec^2$; cot $\rightarrow$ − csc ² with $1 + \cot^2 = \csc^2$.	[T]

[T] Phuc’s difficulty: wrote $\cos x$ with **no sign** for $\frac{d}{dx} \sin x$ and “ $\pm$ ” for $\frac{d}{dx} \cos x$ — “I usually confuse about the plus minus, I don’t remember which one.” Remedy = the $+-+--$ pattern above. **Booklet status [S]:** these derivatives are **not reprinted** in the exam formula booklet — students must know them. That is exactly why Harry drills them.

4. Trig identities — reciprocal / Pythagorean / double-angle

Recall drill **C1-C9**, all **Solved** (some after correction). Harry: “it’s really important that you have all of that memorized.”

4.1 Reciprocal identities (C1-C4) [T]

Identity	Condition	Memory	Src
$\tan A = \frac{\sin A}{\cos A}$	Undefined where $\cos A = 0$ [S] .	Definition.	[T][S]
$\csc A = \frac{1}{\sin A}$	Undefined where $\sin A = 0$ [S] .	Third letter: cosec $\rightarrow$ sin.	[T]
$\sec A = \frac{1}{\cos A}$	Undefined where $\cos A = 0$ [S] .	sec $\rightarrow$ cos.	[T]
$\cot A = \frac{\cos A}{\sin A}$	Undefined where $\sin A = 0$ [S] .	Reciprocal of tan.	[T]

4.2 Pythagorean identities (C5-C7) [T]

Identity	Derivation / meaning	Trap	Src
$\sin^2 A + \cos^2 A = 1$	The master identity.	—	[T][S]
$1 + \tan^2 A = \sec^2 A$	Divide the master by $\cos^2 A$ [S] .	Phuc trap: said $\cot^2 A$. It is $\sec^2 A$.	[T]
$1 + \cot^2 A = \csc^2 A$	Divide the master by $\sin^2 A$ [S] .	—	[T]

4.3 Double-angle identities (C8-C9) [T]

Identity	Note / derivation	Trigger	Trap	Src
$\sin 2A = 2 \sin A \cos A$	Double angle.	Spotting $2 \sin A \cos A$ (e.g. $8 \sin x \cos x$).	Phuc first reached for $\sin(A + B)$.	[T][S]
$\cos 2A = \cos^2 A - \sin^2 A$	Order: cosine first.	Default $\cos 2A$.	Phuc trap: $\sin^2 A - \cos^2 A$ (wrong order); $\cos^2 A - 2 \sin^2 A$.	[T]
$\cos 2A = 1 - 2 \sin^2 A$	Sub $\cos^2 A = 1 - \sin^2 A$ [S] .	Isolating $\sin^2 A$.	—	[T]

Identity	Note / derivation	Trigger	Trap	Src
$\cos 2A = 2 \cos^2 A - 1$	Sub $\sin^2 A = 1 - \cos^2 A$ [S]. Rearranges to $2 \cos^2 A = \cos 2A + 1$.	Used in Q4(a) for $4 \cos^2 x$.	—	[T]

[T] Harry's rule: “memorising these three forms rather than deriving them during the exam” — deriving costs time; “the kind of thing that is really helpful for flashcards.” The three-step derivation (start $\cos^2 A - \sin^2 A \rightarrow$ sub $\cos^2 A = 1 - \sin^2 A \rightarrow 1 - 2 \sin^2 A$; sub $\sin^2 A = 1 - \cos^2 A \rightarrow 2 \cos^2 A - 1$) is kept [S] for understanding only.

5. R-form & solving the trig equation (Q4)

Q4 [T] (from the board — no QP file staged): $f(x) = 8 \sin x \cos x + 4 \cos^2 x - 3$. This is the payoff for the identity drill.

5.1 Q4(a) — to $a \sin 2x + b \cos 2x + c$ • Solved

Step	Working	Identity used	Src
$8 \sin x \cos x$	$= 4(2 \sin x \cos x) = 4 \sin 2x$	$\sin 2A = 2 \sin A \cos A$	[T]
$4 \cos^2 x$	$2 \cos^2 x = \cos 2x + 1 \Rightarrow 4 \cos^2 x = 2 \cos 2x + 2$	$\cos 2A = 2 \cos^2 A - 1$	[T]
Assemble	$f(x) = 4 \sin 2x + (2 \cos 2x + 2) - 3 = 4 \sin 2x + 2 \cos 2x - 1$ $a = 4, b = 2, c = -1$	—	[T]

[T] “Really need to know those trig identities ... if you don't know them, it's very tricky.”

5.2 Q4(b) — to $R \sin(2x + \alpha) + c$, $R > 0$, $0 < \alpha < \frac{\pi}{2}$ • Solved

Method / line	Conditions & meaning	Trap	Src
Expand $R \sin(2x + \alpha) = R \cos \alpha \sin 2x + R \sin \alpha \cos 2x$	Compound-angle identity [S].	Match to $4 \sin 2x + 2 \cos 2x$.	[T][S]
Match: $R \cos \alpha = 4$, $R \sin \alpha = 2$	Equate $\sin 2x$, $\cos 2x$ coefficients.	Line up which is a , which is b .	[T]
R (square-and-add): $R^2 = 4^2 + 2^2 = 20 \Rightarrow R = 2\sqrt{5}$	$R^2(\cos^2 \alpha + \sin^2 \alpha) = a^2 + b^2$; exact surd.	Phuc trap: $R = 4.472$ (decimal). Harry: “Exact, no decimals allowed.”	[T][S]
α : $\tan \alpha = \frac{2}{4} = \frac{1}{2} \Rightarrow \alpha = \arctan \frac{1}{2} \approx 0.464$	Radians , 3 s.f. Quadrant: $a, b > 0 \Rightarrow$ first quadrant [S].	$\tan \alpha$ alone doesn't fix α — the signs of a, b do (atan2).	[T][S]

Method / line	Conditions & meaning	Trap	Src
Assemble	$f(x) = 2\sqrt{5}\sin(2x + \alpha) - 1$, $\alpha \approx 0.464$ ($c = -1$ carries over).	0.464 is a 3-s.f. approximation, not exact.	[T]

5.3 Q4(c) — max value and second smallest x

Part	Working	Trap	Src
(c)(i) max · Solved	$\sin(\cdot)_{\max} = 1 \Rightarrow f_{\max} = 2\sqrt{5}(1) - 1 = 2\sqrt{5} - 1$ (exact).	Keep it exact.	[T]
(c)(ii) 2nd smallest positive x · Solved	Set $\sin(2x + \alpha) = 1$; whole angle $2x + 0.464 = \frac{\pi}{2}, \frac{5\pi}{2}$; $x \approx 0.553, 3.69$. Answer $x \approx 3.69$.	Headline Phuc trap.	[T][S]

[T] The headline mistake of the lesson. Phuc found the first x then added 2π to **that** x . Harry: “ones are two π apart, but these ones, when you’ve rearranged, are not two π apart” — the 2π periodicity belongs to the **whole angle** $2x + \alpha$, before dividing by 2. **Rule:** write all solutions for the whole angle first, then rearrange for x . Also **3 s.f.** $\neq$ **3 d.p.**: $3.694 \rightarrow 3.69$. **[S] general solution:** $\sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2} + 2n\pi$.

6. Transformations & integration applications (Q1, Q2)

Q1 & Q2 [T] (from the board).

6.1 Q1 — modulus & transformations · $f(x) = 2|x - 5| + 10$, P the vertex

Part	Working	Note / trap	Src
(a) P · Solved	Piecewise $2x$ ($x \geq 5$), $-2x + 20$ ($x < 5$); $f(5) = 10 \Rightarrow P = (5, 10)$.	Shortcut: vertex of $a x - h + k$ is (h, k) .	[T]
(b) solve $2 x - 5 + 10 > 6x$ · Solved	$6x$ steeper than $2x$; on the left branch $6x = -2x + 20 \Rightarrow x = \frac{5}{2}$; solution $x < \frac{5}{2}$.	Sketch first to pick the branch. Calculator solutions not acceptable — show algebra.	[T]
(c) image of P under $y = 3f(x - 2)$ · Solved	$x - 2$: right 2 ($5 \rightarrow 7$); $\times 3$: y stretch ($10 \rightarrow 30$); $P' = (7, 30)$.	Inside $\Rightarrow x$; outside $\Rightarrow y$.	[T]

[T] Exam tip Harry stressed 3x: if you can’t do an early part (e.g. finding P), **invent** a plausible value and carry it forward — later parts still earn method marks.

6.2 Q2 — algebraic division, then integration · $g(x) = \frac{2x^2 - 5x + 8}{x - 2}$

Part / method	Working	Trap	Src
(a) M1 long division · Solved	$2x^2 - 5x + 8 = (x - 2)(2x - 1) + 6 \Rightarrow$ $g = 2x - 1 + \frac{6}{x-2};$ $A = 2, B = -1, C = 6.$	Phuc trap: $-5x - (-4x) = -9x$. Harry: “always bracket the term you subtract” $\Rightarrow -x$. Remainder 6 sits over $(x - 2)$ alone.	[T]
(a) M2 substitution · Solved	$2x^2 - 5x + 8 \equiv (Ax + B)(x - 2) + C:$ $x = 2 \Rightarrow C = 6; x = 0 \Rightarrow B = -1;$ $x = 1 \Rightarrow A = 2.$	Both methods agree; Harry prefers division for speed.	[T]
(b) $\int_4^8 g \, dx = \alpha + \beta \ln 3$ · Solved	$[x^2 - x + 6 \ln x - 2]_4^8 = (64 - 8 +$ $6 \ln 6) - (16 - 4 + 6 \ln 2) = 44 + 6 \ln 3;$ $\alpha = 44, \beta = 6.$	Uses $\int \frac{6}{x-2} = 6 \ln x - 2 $; log law $\ln 6 - \ln 2 = \ln 3.$	[T][S]

[S] $f(x) > 0$ / modulus boundary: write $\int \frac{1}{x-2} = \ln |x - 2| + c$ **with the modulus**; the bars drop here only because on $[4, 8]$ the argument $x - 2 > 0$. This positivity condition is the spec-level layer, not a spoken lesson line.

6.3 Q3 — log-log graph · Discussed only (D)

Phuc’s sketch of $\log_{10} y$ against $\log_{10} x$ was “pretty much fine.” Two points only, **no full tutor working reconstructed**: (i) label intercepts as **coordinate pairs**, e.g. $(0, 6)$, $(2, 0)$; (ii) a “**sketch**” needs shape + key points, a “**plot**” needs an accurate scale. [T]

7. Evidence-based Phuc traps & technique boundaries

7.1 Phuc’s own errors → Harry’s corrections (diagnostic) [T]

Topic	Phuc’s attempt / error	Harry’s correction
$\frac{d}{dx}(a^x)$	$x a^{x-1}$ (power rule mis-applied)	Variable in the power $\Rightarrow a^x \ln a$.
$\frac{d}{dx}(\ln 3x)$	$\frac{1}{3x}$ (missing f')	Numerator $f' = 3$: $\frac{3}{3x} = \frac{1}{x}$.
$\frac{d}{dx}(5 \ln 2x)$	$5 \cdot \frac{10}{2x}$	$5 \cdot \frac{2}{2x} = \frac{5}{x}$.
$\frac{d}{dx} \sin x, \frac{d}{dx} \cos x$	$\cos x$ (no sign); “ $\pm$ ”	$+\cos x; -\sin x$; use $+$ $-$ $+$ $-$ $+$ $-$.
$1 + \tan^2 A$	$\cot^2 A$	$\sec^2 A$.
$\cos 2A$	$\sin^2 A - \cos^2 A; \cos^2 A - 2 \sin^2 A$	$\cos^2 A - \sin^2 A$ (cosine first); $2 \cos^2 A - 1$.
Q2(a) division	$-5x - (-4x) = -9x$; remainder mis-written	Bracket what you subtract $\Rightarrow -x$; $6/(x - 2)$.
Q4(b) R	$R = 4.472$ (decimal)	Exact: $R = 2\sqrt{5}$.
Q4(c)(ii)	add 2π to x after rearranging	Add 2π to the whole angle $2x + \alpha$ first .

Topic	Phuc's attempt / error	Harry's correction
Q4(c)(ii) rounding	3.694 (4 s.f.)	3.69 (3 s.f.); s.f. $\neq$ d.p.

[T] Diagnostic read-across: two themes recur — **(i)** distinguishing the power rule from the exponential/log rules (*where the variable sits*), and **(ii)** *finishing carefully* — signs, exact surd form, significant figures, and adding 2π at the right stage. The mathematics is landing; the marks leak at the edges.

7.2 The four explicit boundaries (consolidated)

Boundary	Statement	Tier
$f(x) > 0$	Required for $\frac{d}{dx} \ln f(x) = \frac{f'}{f}$ to be real, and for dropping the modulus in $\ln f(x) $. Validity condition; all lesson examples satisfy it.	[S] (not a spoken line)
Radians	Derivative formulae assume x in radians; Q4(b) <i>demand</i> α in radians (3 s.f.).	Derivative assumption [S] ; Q4 demand [T]
Phase / quadrant	$R \cos \alpha = a$, $R \sin \alpha = b$; α fixed by signs of a, b (atan2), not by $\tan \alpha$ alone. Here $a, b > 0 \Rightarrow$ 1st quadrant.	[S]
Tutor-vs-official	Formulae drilled from memory = [T] ; domain conditions, radian assumption, identity derivations, general $R/\sin \theta = 1$ rules = [S] (spec-derived, not Harry's words).	boundary itself

7.3 Exam-technique boundaries Harry named [T]

Technique	Statement
Exact vs rounded s.f. vs d.p.	Exact = surds/fractions, no decimals; square-and-add for exact $R = 2\sqrt{5}$. Round the final answer to the <i>demand</i> accuracy; $3.694 \rightarrow 3.69$ (3 s.f.).
Whole-angle first	Apply $+2\pi$ to $2x + \alpha$, then divide for x — never 2π onto x .
Modulus in logs	$\ln x - 2 $; remove bars only once the argument is positive.
Calculator-banned	Q1(b): calculator solutions “not acceptable” — show the algebra.
Invent-and-carry	Assume a stuck early value; carry forward for method marks.
Sketch vs plot	Sketch = shape + labelled key points; plot = accurate scale.
Memorise, don't derive	Memorise the three $\cos 2A$ forms and the six derivatives (flashcards).

7.4 Progress / study-plan context (not taught maths) [T]

Kept as context only: on the submitted paper Phuc “scored ... 24 out of 75, which was only 23 UMS ... a U-grade still,” and “a very long way off ... from getting you to the A-star.” (Harry hesitated 23/24 on the recording; raw $\approx 24/75, \approx \$23$ UMS, grade U — the Loom summary’s “about 23/75” is orientation.) He framed the curve encouragingly: early papers go “very badly” because “there’s so much they don’t yet know,” then “lots of things start to click.” Scheduling (readiness, next work = P3 Jan 2024 + revisit M1 Oct 2023) is **out of scope** as non-maths planning.

7.5 Independent-study advice — Harry’s email (email, not video) [E]

Point	Harry’s guidance (key phrasing preserved)
Re-do covered questions	Working them “yourself from scratch”, unaided, is “an effective use of time” — doing a question live with a tutor “can feel quite different” to doing it alone; it lets you “confirm ... that you have indeed learned ... without my support.”
Space your practice	“It is natural that we forget things ... that’s why it’s important that we space out our practice over time, and revisit regularly ... this way we can combat the ‘forgetting’ curve.”
Both have a place	Re-done questions “wouldn’t give you the same degree of insight ... compared to ... an ‘unseen’ paper,” but “there is a place for both.”

8. Sources & status boundaries (compact)

- **Primary — [T] Tutor:** Loom recording, <https://www.loom.com/share/7dc9017098bf4631b50e700640fb9c85>, via the spoken transcript sources/loom_transcript.txt (primary) and the six-segment audio+visual ledger analysis/gemini_video_evidence.md (governing evidence). Tutor **Harry Williams** (Get Me That Grade), student **Phuc Nguyen**, 12 July 2026. **The June 2024 P3 question statements are [T]** — reconstructed from Harry’s shared board; **no official question-paper text file was staged** for this lesson, so the wording is tutor-transcribed, not verbatim-official.
- **Reference — [S] Spec / formula booklet:** *Edexcel IAL Mathematics 2018 specification* (sources/Edexcel_IAL_Maths_2018_specification.txt): 2.1-2.3 (secant/cosec/cot, $\sec^2 \equiv 1 + \tan^2$, $\csc^2 \equiv 1 + \cot^2$, double angle, R -form), 4.1-4.2 (standard + reciprocal-trig derivatives, product/quotient/chain), 4.4 ($\frac{d}{dx}a^x = a^x \ln a$), 5.1-5.2 ($\int \frac{1}{x} = \ln|x|$, $\int \frac{f'}{f} = \ln|f|$), 1.1/1.3/1.4 (division, modulus, transformations). Used **only** for clearly-labelled formal expansions ($a > 0$, $f(x) > 0$, $\cos A \neq 0$, the radian assumption, identity derivations, general $R/\sin \theta = 1$ rules); never voiced as Harry.
- **[E] Email:** sources/harry_followup_email_2026-07-12.txt (12 Jul 2026, Harry→Phuc) — spaced-practice / independent-study advice, §7.5. Email, not video.
- **Cross-check only:** SaveMyExams pages (sources/savemyexams_double_angle.html, _exponential_derivatives.html, _reciprocal_trig_derivatives.html, _trig_further_identities.html) — method sanity checks, never allowed to overwrite tutor / spec evidence.

Status rules (load-bearing): warm-up A1-A19, trig table B1-B8, identities C1-C9, Q1(a-c), Q2(a-b), Q4(a-c) = **Solved**; Q3 = **discussed only** (no working reconstructed); progress/scheduling = context only (§7.4). **Booklet status [S]:** the trig identities and standard derivatives are results students must *know* — they are **not reprinted** in the exam formula booklet, which is why Harry drills them. This pack does **not** export to Notion, publish, or claim downstream release — the parent verifies and exports.